

Mohanbalaji Devarapu | Data Analyst

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SUMMARY

Data Analyst with 3+ years of experience delivering data-driven solutions across healthcare, insurance, and software industries in the US and India. Skilled in SQL, Python, R, and advanced analytics frameworks (Pandas, NumPy, Scikit-learn, SHAP) with proven expertise in building predictive models, optimizing ETL/ELT pipelines (dbt, Apache Airflow, Kafka, Snowflake, BigQuery), and standardizing reporting frameworks. Adept at designing interactive dashboards in Power BI, Tableau, and Looker Studio to communicate insights that drive measurable cost reduction and process improvement. Experienced in collaborating with cross-functional teams including clinicians, actuaries, product managers, and operations leaders to translate business goals into measurable KPIs. Strong background in regulatory compliance (HIPAA, IRDAI) and data governance, with a track record of reducing reporting errors, improving customer retention, and enabling executive decision-making through scalable analytics solutions.

TECHNICAL SKILLS

Languages:	SQL, Python, R, DAX, Power Query (M), Bash
Frameworks & Libraries:	Pandas, NumPy, Scikit-learn, SHAP, Matplotlib, Seaborn, ggplot2, dplyr, tidyverse, dbt
Database:	SQL Server, MySQL, PostgreSQL, Snowflake, Google BigQuery, MongoDB
Visualization Tools:	Power BI, Tableau, Looker Studio, Excel (Pivot Tables, Power Query), Canva
Tools:	Excel, Jupyter Notebook, VS Code, Git, Jira, Confluence, Google Sheets, SharePoint, Notion
Data Warehousing & Pipelines:	dbt, Apache Airflow, Snowflake, Redshift, BigQuery, Kafka, ETL/ELT Pipelines
Project Management & Collaboration:	Agile/Scrum, Collibra (Data Governance), Role-Based Access Control (RBAC), Metadata Management, Audit Logs, HIPAA, IRDAI Regulatory Compliance

EXPERIENCE

Data Analyst | Humana Inc., KS

Jun 2025 – Present

- Consolidated medical claims, pharmacy, and EHR datasets into Snowflake, documenting data dictionaries and lineage, which reduced metric inconsistencies across business units by 28%.
- Partnered with clinicians, actuaries, and operations leaders to define utilization KPIs (ER visits, readmission rates, drug adherence), mapping each KPI to specific claims and provider feeds.
- Automated ETL workflows with dbt + Apache Airflow, transforming raw claim transactions into member-level cost & utilization fact tables, achieving hourly refresh for real-time analytics.
- Built predictive cost-risk models in Python (Scikit-learn, SHAP) to identify members likely to incur avoidable hospitalizations, enabling early outreach that reduced preventable admissions by 11%.
- Performed quality audits in SQL Server to validate claims vs EHR encounters, detecting and correcting mismatches worth \$2.3M in reporting errors.
- Designed Power BI dashboards that visualized ER utilization, pharmacy spend, and readmission trends, empowering executives to implement interventions that lowered total medical cost by 7% YoY.

Data Analyst Intern | Humana Inc., KS

Mar 2025 – May 2025

- Documented source-to-target mappings in Confluence, capturing data fields for claims, pharmacy, and EHR feeds to support downstream analytics and ensure traceability.
- Profiled raw claims datasets using Excel Power Query, identifying duplicates and incomplete provider IDs that caused 12% delays in claims adjudication.
- Assisted in staging claims data pipelines by developing SQL scripts for initial data extraction and validation checks, ensuring completeness before loading into Snowflake.
- Conducted exploratory analysis in R (ggplot2, dplyr) to uncover utilization patterns across demographics, surfacing insights like higher ER dependency in Tier-3 regions.
- Supported pilot testing of readmission risk models by validating training and test data splits in BigQuery, ensuring statistical balance across cohorts.
- Built Tableau reports summarizing hospital utilization and cost trends, which were presented in monthly stakeholder reviews to guide resource allocation.

Data Analyst | Adons Softech, India

Jun 2022 – Jun 2023

- Standardized claims lifecycle data models using dbt & SQL, enforcing consistent metric definitions across underwriting, operations, and finance teams, and reducing reporting discrepancies by 30%.
- Interpreted claim rejection trends through historical denial analysis in Python (Pandas, Seaborn), delivering actionable insights that improved first-pass claim approvals by 17%.
- Audited premium payments & claim disbursement logs with SQL Server & Excel reconciliation routines, identifying mismatches worth ₹1.2M and accelerating financial closure with the finance team.
- Explored policy lapse patterns by region and demographics using R (ggplot2, dplyr), providing data-driven inputs that improved renewal rates in Tier-2 regions by 12%.
- Built Power BI dashboards to track turnaround time (TAT) across claim categories, enabling the operations team to detect approval bottlenecks and shorten processing cycles by 20%.
- Delivered executive-level insights via Looker Studio and Canva presentations, and maintained compliance datasets with SharePoint audit trails, ensuring IRDAI regulatory adherence and risk-free quarterly reporting.

Data Analyst Tophat Software Technologies, India	Jan 2021 – May 2022
- Standardized event taxonomy and KPI definitions, creating a centralized data dictionary that improved reporting accuracy by 35% and reduced duplicated metrics across teams. - Translated stakeholder objectives into measurable KPIs, aligning 10+ product/marketing goals with reliable data sources and reducing ad-hoc reporting requests by 25%. - Engineered real-time ETL pipelines using Kafka, AWS S3, and Snowflake, achieving <60s data freshness and enabling faster decision-making for product and campaign teams. - Performed cohort, funnel, and anomaly analyses with SQL/Python, uncovering insights that increased user retention by 12% and reduced missed defect detection by 30%. - Designed and validated A/B experiments, leveraging statistical testing to recommend campaign and feature rollouts that boosted conversion rates by 15% and optimized marketing ROI. - Built interactive Power BI dashboards with automated data refresh and alerting, cutting reporting turnaround time by 40% and improving incident response times by 20%.	

EDUCATION	
Master of Science in Information Studies	May 2025
Trine University, Indiana	
Bachelor in Computer Science & Engineering	Jun 2022
QIS College of Engineering and Technology, India	

PROJECTS	
SAP Ariba Buying & Procurement Process Optimization	
- Streamlined the SAP Ariba procurement workflow for the American Retail team at Wells Fargo. - Successfully coordinated purchase orders, goods delivery, installation, and vendor engagement processes. - Enhanced procurement tracking and operational transparency using ServiceNow and EPRO. - Delivered consistent technical support and process improvements, ensuring alignment with organizational standards.	
Technical Support and Service Quality Enhancement for Wells Fargo Advisors	
- Provided technical support and ensured service quality maintenance for the Wells Fargo Advisors team in the USA. - Utilized tools such as ServiceNow and Wanda to effectively manage and resolve technical issues. - Maintained high standards of service delivery by validating and addressing requests with accuracy.	